

Generation Interconnection System Impact Study Report

For Queue Project Q11-0006

Gisler 138 kV
February 17, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0006. The Interconnection Customer proposes a 14.5 MW generating facility interconnecting at the Gisler 138 kV substation in the Pennsylvania area. The request was received on January 22, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 4 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,480,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wallenpaupack 500-345 kV Transformer
Contingency: Clear Springs 500/345 kV TX
Loading: 110.0% (Rating: 910 MVA)
DFAX: 5.0%
Pre-project loading: 110.0%
2. Facility: Pok St-Van Wagner 138 KV L
Contingency: Walnut-W Gallatin 345 KV
Loading: 107.3% (Rating: 242 MVA)
DFAX: 9.9%
Pre-project loading: 106.7%
3. Facility: Verde Mar 2605-Croton Falls 138 KV Line
Contingency: West Gallatin-Walnut 345 KV
Loading: 106.8% (Rating: 270 MVA)
DFAX: 7.6%
Pre-project loading: 106.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wallenpaupack 500-345 kV Transformer	Replace transformer with higher-rated unit	\$26,254,000	100% (\$26,254,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Polk St-Van Wagner 138 KV	Reconductor line	\$3,030,000	100% (\$3,030,000)
Verde Mar 2605-Croton Falls 138 KV Line	Reconductor line	\$2,196,000	100% (\$2,196,000)
	Total allocated to project		\$31,480,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.